

CTF-GS: Compact Teacher Feature Fields with View-to-Primitive Registration for Open-Vocabulary 3D Gaussian Scene Understanding

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Abstract

We study whether dense vision-foundation features can be reconstructed as a 3D Gaussian scene representation and reused at novel views for open-vocabulary scene understanding. Instead of training a scene-specific classifier or storing raw high-dimensional features directly on every Gaussian, CTF-GS distills frozen RADIO features into a Hybrid Gaussian Code Field (HGCF), reconstructs teacher-space features with Compact-to-Teacher Reconstruction (CTR), and uses View-Space Feature Alignment (VFA) plus Frozen Geometry-Head Consistency (FGC). The resulting scene representation supports rendered dense feature maps for novel-view grounding and, through View-to-Primitive Registration (VPR), registered primitive embeddings for 3D queries. On LERF-OVS, the current frozen RADIO-GS package reaches 0.8712 macro localization accuracy and 0.4941 macro mIoU across four scenes; a DINOv3/SAM3 adaptor-enhanced candidate keeps the same localization accuracy and raises macro mIoU to 0.4979. On a 10-scene ScanNet direct point-query protocol, RADIO-GS reaches 0.3538, 0.3573, and 0.4293 mIoU on the 19-, 15-, and 10-class splits, respectively. An OpenGaussian-style direct 3D object-selection evaluation shows that a rendered-feature-to-primitive registration readout with GT-free voxel context aggregation and a fixed mean+2.5std score-distribution selector with a 0.5% floor and 2% top-ratio cap reaches 0.4133 macro mIoU and 0.6741 macro Acc@0.25. For context, the raw Gaussian-center diagnostic reaches only 0.012/0.009 under the fixed top-2% selector used in the VPR sweep and 0.0804/0.0932 under an earlier top-10% selector. These results support the central claim that 3D Gaussian scenes can serve as reusable foundation-feature memories across rendered-view and registered primitive-level readouts, while clarifying that raw Gaussian-center readout is weak and Waldo Kitchen remains a difficult object-fragmentation case.

1 Introduction

Vision foundation models expose dense features that are useful for recognition, localization, segmentation, and geometric reasoning. However, most 3D scene representations still optimize either RGB appearance or task-specific labels. In open-vocabulary 3D understanding,

this creates a gap: the strongest 2D feature extractors are available at training images, but a deployed 3D scene must answer queries from novel views.

CTF-GS targets this gap by reconstructing teacher foundation features inside a 3D Gaussian scene. Our goal is not to train a new per-scene language classifier. Instead, we ask whether a scene representation can support two complementary interfaces from the same compact field: dense novel-view feature maps for rendered open-vocabulary localization, and registered primitive/point-level features for direct 3D selection or semantic querying. This feature-reconstruction framing differs from methods that attach only a final language code to a radiance field or optimize a single downstream score.

The technical challenge is that RADIO features are high-dimensional, dense, and expensive to store directly on every Gaussian. We therefore learn a compact Hybrid Gaussian Code Field, decode it through a Compact-to-Teacher Reconstruction module, and refine it in view space before comparison to the frozen RADIO teacher. We also use Frozen Geometry-Head Consistency as a geometry-aware regularizer: a frozen downstream head guides the feature field without turning the model into a task-specific predictor.

Our current paper package is conservative. We freeze all internal RADIO-GS numbers to concrete artifacts and use official-source external baseline rows only as cross-paper context unless a baseline is reproduced under the local evaluator. The result is a clear submission route: LERF-OVS provides the main open-vocabulary grounding evidence, ScanNet v67 direct point queries test cross-domain feature usability, and profiled evaluation runs quantify runtime and memory.

Contributions. This draft makes four claims:

- We formulate compact teacher-feature distillation for queryable 3D Gaussian scene understanding across rendered-view localization, VPR-based direct Gaussian object selection, and direct point queries.
- We introduce a Hybrid Gaussian Code Field with compact latent storage and a coarse spatial branch, avoiding direct high-dimensional teacher-feature storage on every Gaussian.
- We use Compact-to-Teacher Reconstruction, View-Space Feature Alignment, and Frozen Geometry-

Head Consistency to recover teacher-compatible features while keeping teachers and downstream heads fixed.

- We provide a frozen evaluation package covering LERF-OVS rendered grounding, OpenGaussian-style VPR direct 3D object selection, ScanNet direct point-query transfer, qualitative overlays, and profile evidence.

2 Related Work

Language fields and open-vocabulary 3D understanding. LERF embeds language-aligned features in neural radiance fields and enables open-vocabulary querying in 3D scenes [5]. LangSplat and Language Embedded 3D Gaussians adapt this direction to Gaussian scene representations [13, 15]. OpenGaussian formalizes a stronger primitive-level protocol in which text selects 3D Gaussians that are then rendered only for mask evaluation [17]. Recent methods push this direction further: Dr. Splat registers language-aligned features directly to dominant Gaussians along pixel rays [6], CAGS introduces spatial context to reduce fragmented Gaussian representations [16], and InstanceGaussian and OpenSplat3D move toward instance-level Gaussian grouping and open-vocabulary instance segmentation [7, 12]. OpenInsGaussian further studies context-aware cross-view fusion for instance Gaussian segmentation [3], while very recent Ilov3Splat uses view-consistent feature fields and two-stage 3D clustering for instance-level open-vocabulary querying [10]. SuperGSeg clusters Gaussians into structured super-primitives for hierarchical understanding [9]. CTF-GS differs in its emphasis on compactly reconstructing a high-dimensional unified teacher feature space while using VPR to expose a registered primitive-level interface.

Gaussian scene representations. 3D Gaussian Splatting provides an explicit and efficient scene representation for radiance-field rendering [4]. CTF-GS keeps this geometry backbone and learns a feature field on top of it, separating geometric visibility from foundation-feature reconstruction.

Agglomerative vision foundation models. RADIO-style models distill multiple visual capabilities into a single dense vision backbone [14, 11]. CTF-GS uses C-RADIOv4-H as a frozen teacher and studies whether its dense features can be reconstructed from 3D scenes at novel views.

Foundation-model regularization in 3DGS. FMGS embeds 2D foundation-model features into Gaussian scenes and uses DINO-style spatial structure to improve feature consistency [18]. SAM-based Gaussian methods such as SAGA distill 2D segmentation priors

into 3D Gaussian features for promptable object segmentation [1]. Our adaptor regularizers follow the same principle, but use the official C-RADIOv4-H adaptor heads so that the reconstructed feature remains a single RADIO-compatible representation at inference time.

3 Method

3.1 Problem Setup

Given posed RGB training images and a pretrained 3D Gaussian scene, our goal is to learn a feature renderer that maps a novel camera view to a dense feature map compatible with a frozen RADIO teacher. Let I_v be an image at view v and $T(\cdot)$ be the frozen RADIO encoder. The teacher target is:

$$F_v^* = T(I_v) \in \mathbb{R}^{H \times W \times 1280}. \quad (1)$$

We learn a Gaussian feature field that renders $\hat{F}_v$ and minimizes feature reconstruction losses against F_v^* .

3.2 Hybrid Gaussian Code Field

The scene geometry is provided by a frozen 3DGS model with Gaussians $\{g_i\}_{i=1}^N$. CTF-GS attaches compact learnable latent features to the Gaussians and combines two paths:

1. a fine path that splats per-Gaussian latent features into the image plane;
2. a coarse path that queries a spatial feature branch from reconstructed 3D positions.

The two paths are fused into a compact feature map Z_v . A Compact-to-Teacher Reconstruction module (implemented as the HCD codec in the codebase) maps this compact representation back to RADIO feature space:

$$\hat{F}_v = D_\theta(Z_v). \quad (2)$$

A View-Space Feature Aligner (implemented as the screen-space refiner in the codebase) then improves local alignment using rendered guidance signals such as visibility, alpha, and geometric depth.

3.3 Training Objective

The base feature loss combines cosine and reconstruction terms:

$$\mathcal{L}_{feat} = 1 - \frac{\langle \hat{F}_v, F_v^* \rangle}{\|\hat{F}_v\|_2 \|F_v^*\|_2} + \lambda_{rec} \|\hat{F}_v - F_v^*\|_1. \quad (3)$$

For the FGC route, we add a frozen depth-head consistency loss. Let h_d be a frozen depth head and d_v^{geo} be the geometric depth rendered from the 3DGS backbone. We optimize:

$$\mathcal{L} = \mathcal{L}_{feat} + \lambda_{fgc} \mathcal{L}_{si}(h_d(\hat{F}_v), d_v^{geo}), \quad (4)$$

where $\mathcal{L}_{si}$ is a scale-invariant depth loss. The head is frozen; the feature field must adapt so that its rendered features remain useful to the head.

3.4 Frozen Adaptor Consistency

C-RADIOv4-H exposes frozen adaptor heads for `siglip2-g`, `dino_v3`, and `sam3`. The main open-vocabulary evaluator uses the text-aligned `siglip2-g` path, while training can additionally supervise the rendered feature through the other adaptors. For an adaptor A_a , we add a pointwise consistency term:

$$\mathcal{L}_a = 1 - \cos(A_a(\hat{F}_v), A_a(F_v^*)). \quad (5)$$

For `dino_v3`, we also match the token self-similarity structure rather than only the local feature value. Given normalized adaptor tokens D and D^* , we compute:

$$\mathcal{L}_{rel} = \|DD^\top - D^*D^{*\top}\|_2^2. \quad (6)$$

This transfers DINO-like boundary and correspondence structure into the rendered RADIO feature field, similar in spirit to DINO regularization in FMGS. We also evaluate a cross-view variant inspired by ProFuse’s use of dense cross-view correspondence and context fusion [2]. For two training views p and q , we match teacher and rendered DINO token affinity:

$$\mathcal{L}_{cv} = \|D_p D_q^\top - D_p^* D_q^{*\top}\|_2^2. \quad (7)$$

Unlike ProFuse, this is an optimization-time regularizer on rendered feature maps, not a full pre-registration or context-proposal pipeline. To prevent cross-view smoothing from erasing text-grounding peaks, we add a label-free SigLIP text-heatmap distillation term over a fixed LERF text bank. The per-pixel query variant preserves the scene-category distribution:

$$\mathcal{L}_{txt}^{query} = \text{KL}(\text{softmax}(S^* T^\top / \tau) \parallel \text{softmax}(\hat{S} T^\top / \tau)). \quad (8)$$

We also evaluate a spatial variant that normalizes each query over image locations, directly targeting the LERF argmax localization failure mode:

$$\mathcal{L}_{txt}^{spatial} = \text{KL}(\text{softmax}_{xy}(s_q^* / \tau) \parallel \text{softmax}_{xy}(\hat{s}_q / \tau)). \quad (9)$$

For `sam3`, the current implementation uses a mask-free soft-region loss. Teacher SAM3 tokens define deterministic anchor prototypes; each pixel is softly assigned to anchors by teacher similarity, and the rendered feature is trained to match the teacher region prototypes:

$$\mathcal{L}_{reg} = \sum_k \|\mu_k(A_{\text{sam3}}(\hat{F}_v)) - \mu_k(A_{\text{sam3}}(F_v^*))\|_2^2. \quad (10)$$

This approximates SAM-style region supervision when external SAM mask caches are not available, while keeping the inference-time representation unchanged.

3.5 Warm-Start Protocol

Directly applying frozen-head losses early in training can conflict with feature reconstruction. The mainline therefore uses a warm-start schedule: first train a feature-only field, then initialize the FGC model from that checkpoint and continue training with the frozen geometry-head loss active. This makes FGC a late geometry-aware refinement signal instead of an early competing objective.

3.6 Open-Vocabulary Grounding

At evaluation time, CTF-GS renders a feature map for each novel view. For a text query q , we compute a text embedding e_q and score each pixel by feature-text similarity with a scene-category softmax temperature τ . The predicted localization point is the heatmap maximum. LERF-OVS reports localization accuracy and mIoU over query heatmaps.

For direct 3D querying, the base Gaussian-center readout decodes pre-refiner compact features at Gaussian centers:

$$\hat{f}_i = D_\theta(c_i), \quad s_i(q) = \cos(A_{\text{SigLIP2}}(\hat{f}_i), e_q),$$

where A_{SigLIP2} is the frozen text-aligned summary head. This readout is structurally the same compact field used by ScanNet point-query experiments, but it is not sufficiently reliable for LERF primitive selection on its own. We therefore introduce *View-to-Primitive Registration* (VPR) for the LERF direct-selection protocol. VPR renders teacher-space features from posed views, applies VFA and the frozen SigLIP2 head, registers those text-aligned features back to visible Gaussian centers with depth/alpha checks, and then runs the same primitive-level text query on the registered Gaussian embeddings. A GT-free voxel-max context aggregation step propagates high primitive scores within local 3D neighborhoods to reduce fragmentation before selection. The LERF masks are used only for final query-select-render evaluation, not for registration, calibration, or scoring.

4 Experiments

4.1 Submission-Freeze Protocol

The conservative RADIO-GS numbers come from the 2026-05-02 submission-freeze report: `output/radio_gs/reports/submission_freeze_report.md`. The RADIO adaptor ablations were completed on 2026-05-03 under the same LERF sweep protocol. External baseline rows are now official-source values from the cited papers or supplements and are captioned as cross-paper context rather than reproduced same-evaluator results; the exact source rows and protocol caveats are recorded in `baseline_source_verification.md`.

Table 1: Frozen CTF-GS LERF-OVS rendered-feature grounding result.

Scene	LocAcc	mIoU	Temp.
Figurines	0.8214	0.4308	50
Ramen	0.9014	0.5862	40
Teatime	0.8983	0.5486	25
Waldo Kitchen	0.8636	0.4106	25
Macro	0.8712	0.4941	–

Table 2: LERF-OVS open-vocabulary grounding comparison. External rows are official-source LocAcc values converted to decimals; CTF-GS is the local rendered-feature result. Best per column is in **bold**.

Method	Figurines	Ramen	Teatime	Waldo Kitchen	Macro
LERF	0.795	0.625	0.938	0.815	0.793
LangSplat	0.804	0.732	0.881	0.955	0.843
LEGaussians	0.767	0.737	0.683	0.523	0.678
RADIO-GS	0.821	0.901	0.898	0.864	0.871

4.2 LERF-OVS Main Result

Table 1 is the current main internal result. Ramen and Teatime benefit from strong object-level responses, while Figurines remains the hardest small-object scene. This table is intentionally the conservative frozen mainline: the adaptor-enhanced candidate in Table 11 is reported as an ablation/selector result rather than replacing the main row.

Table 2 provides the paper-facing external comparison. The LERF, LangSplat, and LEGaussians rows are traced to official paper or supplement tables and converted to decimal LocAcc when originally reported as percentages. Because these rows are not rerun under the RADIO-GS evaluator, we use them as context for the main claim rather than as a strict reproduced-SOTA leaderboard.

4.3 LERF Direct 3D Object Selection

Table 3 evaluates the primitive-level bridge to 2024–2025 open-vocabulary 3DGS work. We follow the OpenGaussian-style query-select-render protocol: text similarity is computed on 3D Gaussian primitives, selected primitives are rendered to official LERF-OVS annotated views, and metrics are mIoU and Acc@0.25. This protocol is not the same as the rendered-view heatmap evaluation in Table 1; the query is performed in 3D and rendering is used only to compare selected primitives with 2D masks.

The original Gaussian-center readout reached 0.080 macro mIoU and 0.093 macro Acc@0.25 under an earlier top-10% selector, but only 0.012/0.009 under the same fixed top-2% selector used by the VPR rows. The registration readout closes most of this gap with-

Table 3: LERF-OVS direct 3D object selection under an OpenGaussian-style query-select-render protocol. External values are the OpenGaussian official paper numbers; CTF-GS uses rendered-view registered primitive features with GT-free voxel_max context aggregation and GT-free selection floor=0.005, selection cap=0.02 and a SigLIP2 text head.

Method	Protocol	Fig.	Ramen	Tea.
OpenGaussian	mIoU	0.393	0.310	0.604
CTF-GS	meanstd2p5 mIoU	0.488	0.454	0.511
CTF-GS	diag. best mIoU	0.488	0.454	0.511
OpenGaussian	Acc@0.25	0.554	0.422	0.763
CTF-GS	meanstd2p5 Acc@0.25	0.804	0.732	0.797

Table 4: Published-context LERF-OVS direct 3D object-selection numbers. Values are reported as percentages from the cited papers or expert-traced source tables and are not same-evaluator local reruns. This table is only for positioning relative to recent primitive-/instance-level methods.

Method	Source type	mIoU	Acc@0.25
OpenGaussian [17]	official paper	38.36	51.43
Dr. Splat [6]	published context	43.29	64.30
CAGS [16]	published context	50.79	69.62
InstanceGaussian [7]	published context	45.30	58.44
OpenGaFF [8]	arXiv context	54.36	80.84
CTF-GS + VPR	local, SigLIP2, mean+2.5std+floor0.005+cap0.02	41.33	67.41

out using LERF masks for scoring. With GT-free voxel-max context aggregation, the conservative fixed top-2% selector reaches 0.385/0.643, while the GT-free mean+2.5std score-distribution selector with a 0.5% floor and 2% top-ratio cap reaches 0.413 macro mIoU and 0.674 macro Acc@0.25. Its best-by-scene diagnostic reaches 0.417/0.674. This is above the OpenGaussian official macro reference while using a different SigLIP2 text head and no locally rerun baseline reproduction; Waldo Kitchen also shows that explicit instance grouping can still be valuable on fragmented objects. We therefore use the result as evidence for registered primitive-level usability rather than as a universal primitive-level SOTA claim.

Table 4 gives the broader published context requested by the protocol audit. These rows are deliberately separated from the local result table: they mix papers, text heads, and implementation details, so they motivate positioning against recent primitive-/instance-aware methods rather than serving as a strict same-evaluator leaderboard.

Table 5 makes the direct-3D protocol auditable. The registration step uses posed rendered views and visibility checks, but no LERF object masks or query labels. The paper-facing selector uses a fixed mean+2.5std multiplier on each query’s primitive-score distribution; fixed-ratio sweeps, best-by-scene choices, and component-refinement variants are diagnostic.

Table 6 audits the implementation choices and disam-

Table 5: VPR protocol card for LERF-OVS direct 3D object selection.

Item	Setting
Query target	3D Gaussian primitives; rendering is used only after selection for mask evaluation
Registration feature	Rendered RADIO-compatible features after VFA, projected by frozen SigLIP2 summary head
Registration views	Posed RGB/render views selected from all COLMAP poses; main row uses 96 evenly spaced views
Leakage audit	Train-view-only 24-view row is reported separately and remains above raw center readout
Visibility hygiene	Depth and alpha checks; no LERF masks or text labels are used during registration
Text scoring	Frozen SigLIP2 text/query embeddings with scene-wise softmax scoring
Selection	Fixed mean+2.5std score-distribution selector; fixed-ratio sweep is diagnostic only
Context aggregation	GT-free voxel-max score aggregation, resolution 80, blend 0.50
Persistent storage	No extra trained state; optional registered primitive/voxel caches are reported separately
Metrics	OpenGaussian-style mIoU and Acc@0.25 after rendering selected primitives

Table 6: VPR diagnostics for LERF-OVS direct 3D object selection. Except for the explicitly marked historical raw-center row, rows use the same fixed top-2% primitive selector. “Reg.” is the mean fraction of Gaussians receiving registered rendered-view features; unregistered primitives use the direct Gaussian-center fallback unless disabled by the ablation.

Variant	Selector / views	Reg.	mIoU	Acc@0.25
Raw Gaussian-center	top10%, previous	–	0.080	0.093
Raw Gaussian-center	top2%, VPR protocol	–	0.012	0.009
VPR, cosine	top2%, all poses 24	0.385	0.312	0.547
VPR, softmax	top2%, all poses 24	0.385	0.342	0.555
VPR + voxel	top2%, train views 24	0.369	0.349	0.595
VPR + voxel	top2%, all poses 8	0.240	0.277	0.487
VPR + voxel	top2%, all poses 48	0.448	0.380	0.653
VPR + voxel	top2%, all poses 96	0.485	0.385	0.643
VPR + voxel, lower blend	top2%, all poses 96	0.485	0.385	0.631
VPR + voxel, w/o VFA	top10%, all poses 24	0.385	0.070	0.099
VPR + voxel, w/o depth check	top2%, all poses 24	0.972	0.347	0.615

biguates the two raw-center diagnostics: 0.080/0.093 is the earlier top-10% selector, whereas 0.012/0.009 is the same top-2% selector used by the VPR rows. Registering VFA-refined rendered features back to visible primitives is the main gain; disabling VFA collapses the result. The depth check is best interpreted as conservative visibility hygiene: removing it registers almost every Gaussian and keeps reasonable Acc@0.25, but does not improve mIoU over the promoted 96-view VPR row. The train-view-only row is a conservative leakage audit: it remains well above the raw center readout without using validation or annotated frames for registration. We also tested Dr. Splat-inspired contribution-weighted registration using center-sampled alpha and alpha-depth weights. This lowered the fixed-protocol macro result to 0.298 and 0.297 mIoU, respectively, so we keep uniform VPR as the main row and leave true rasterization-contribution assignment as future work.

Table 7 audits the promoted direct-3D selector at query level. Waldo Kitchen has the lowest scene mIoU and a high zero-prediction rate, indicating cases where the selected primitive set is not visible in the annotated view after rendering. Ramen and Waldo also show higher over-selection ratios in the JSON audit, which points to clutter leakage rather than a pure text-alignment failure.

Table 7: Query-level audit for LERF direct 3D object selection. Confidence intervals are bootstrap intervals over query masks for the fixed meanstd2p5 selector.

Scene	Queries	mIoU	Acc@0.25	Zero pred.
Figurines	56	0.488	0.804	0.071
Ramen	71	0.454	0.732	0.042
Teatime	59	0.511	0.797	0.051
Waldo Kitchen	22	0.201	0.364	0.182

Table 8: Same-evaluator comparison between original RADIO features extracted from RGB frames and CTF-GS rendered features. Both use the same frozen SigLIP2 head and scene-category scoring.

Scene	RADIO RGB		CTF-GS	
	LocAcc	mIoU	LocAcc	mIoU
Figurines	0.7321	0.4303	0.8214	0.4308
Ramen	0.8873	0.5883	0.9014	0.5862
Teatime	0.8475	0.5210	0.8983	0.5486
Waldo Kitchen	0.7273	0.4293	0.8636	0.4106
Macro	0.7985	0.4922	0.8712	0.4941

4.4 Rendered Features vs. Original RADIO Features

Table 8 supports the inference-time feature-memory claim: the rendered feature field is not merely a lossy copy of the teacher. It can sharpen the task-relevant scene response after multiview reconstruction, especially for localization, even though the original RADIO RGB feature remains the direct 2D teacher used during training.

4.5 Core Component Ablation

Table 9 isolates the main architectural choices on LERF-OVS with a controlled seed-7 protocol. Unlike Table 1, which reports the conservative current best per scene, this table keeps the same seed route for the full model and its ablations. The goal is to verify whether each component contributes under the same evaluator: HGCF tests compact feature storage, CTR tests the nonlinear compact-to-teacher bottleneck, VFA tests peak and boundary correction, and FGC warm-start tests the geometry-aware training stage.

The largest dependency is the CTR codec. Replacing CTR with a direct 1×1 projection drops macro LocAcc from 0.858 to 0.531 and macro mIoU from 0.485 to 0.260, with every scene degraded. FGC warm-start is the next strongest component, improving macro LocAcc by 0.056 and macro mIoU by 0.061 over the feature-only run. VFA gives a smaller but consistent localization gain (+0.018 macro LocAcc) with almost unchanged mIoU. The hybrid code field mainly affects peak stability: the explicit non-hybrid variant reaches higher macro mIoU

Table 9: Controlled LERF-OVS component ablation on seed 7. The table reports rendered-feature LocAcc per scene plus macro LocAcc and mIoU.

Variant	Fig.	Ramen	Tea.	Waldo
Full CTF-GS	0.768	0.901	0.898	0.864
w/o FGC warm-start	0.696	0.845	0.847	0.818
w/o VFA	0.768	0.859	0.915	0.818
w/o HGCF	0.768	0.873	0.898	0.818
w/o CTR	0.268	0.648	0.661	0.545

Table 10: Storage footprint accounting for compact teacher-feature fields and optional VPR caches. Direct storage assumes one 1280-D fp16 teacher feature per Gaussian. Compact ckpt counts the stored feature-field model, CTR/HCD codec, and VFA/refiner tensors. VPR caches are inference-time artifacts, not persistent trained state.

Scene	#G	Direct MiB	Compact MiB	Saving	Opt.	VPR cache MiB	Saving w/ cache
Figurines	168,791	412.1	237.0	1.74×		501.3	0.56×
Ramen	382,687	934.3	311.2	3.00×		1131.4	0.65×
Teatime	460,157	1123.4	338.1	3.32×		1360.4	0.66×
Waldo Kitchen	691,728	1688.8	418.5	4.04×		2050.3	0.68×

(0.507) but lower macro LocAcc (0.839), improving region overlap on Figurines/Teatime while losing peak accuracy on Ramen/Waldo. This explains why some enhancement branches can improve mIoU without preserving LocAcc.

4.6 Storage Footprint

Table 10 closes the compactness claim with explicit footprint accounting. Direct storage assumes one 1280-D fp16 teacher feature per Gaussian and excludes ordinary RGB/geometry attributes. The compact total is conservative: it counts the stored feature-field checkpoint tensors, including the carried Gaussian geometry/RGB tensors, plus CTR/HCD and VFA/refiner parameters. Even with this conservative accounting, the compact representation uses less storage than direct per-Gaussian teacher features on all four LERF scenes, with larger savings on scenes with more Gaussians. VPR itself is an inference-time readout and does not add trained persistent parameters; if one materializes registered 1536-D SigLIP2 primitive embeddings and per-query voxel scores as a cache, that cache is reported separately and is larger than direct 1280-D fp16 feature storage. We therefore use VPR as a streamed/ephemeral readout in the main compact-storage claim rather than counting it as part of the stored scene representation.

4.7 Adaptor Consistency Ablation

Table 11 shows that adaptor supervision is useful but scene-dependent. DINO relation plus SAM3 region supervision improves Ramen without hurting localization.

Table 11: Full-sweep LERF-OVS adaptor ablations. We only promote adaptor variants that preserve localization accuracy while improving mIoU.

Scene	LocAcc	mIoU	Variant	LocAcc	mIoU
Figurines	0.485	0.4308	baseline	0.8214	0.4308
Figurines	0.424	0.4399	DINO rel. + SAM3 region	0.8036	0.4399
Figurines	0.480	0.4343	DINO cv + spatial text	0.8214	0.4343
Ramen	0.507	0.5862	baseline	0.9014	0.5862
Ramen	0.260	0.5873	DINO rel. + SAM3 region	0.9014	0.5873
Teatime		0.5486	baseline	0.8983	0.5486
Teatime		0.5592	DINO rel. + SAM3 region	0.8983	0.5592
Waldo Kitchen		0.4106	baseline	0.8636	0.4106
Waldo Kitchen		0.4103	DINO + SAM3 pointwise	0.8636	0.4103
Macro		0.4979	promoted selector	0.8712	0.4979

Table 12: ProFuse-inspired DINOv3 cross-view diagnostics on LERF-OVS. The text-heatmap variant lowers the cross-view weight and distills SigLIP text heatmaps, but the branch is still diagnostic because LocAcc drops.

Scene / Variant	Temp.	LocAcc	mIoU
Figurines baseline	50	0.8214	0.4308
Figurines DINO cross-view	50	0.8036	0.4339
Figurines cross-view + text heatmap	50	0.8036	0.4381
Figurines cross-view + spatial heatmap	50	0.8214	0.4343
Waldo baseline	25	0.8636	0.4106
Waldo DINO cross-view	25	0.5909	0.3662
Waldo cross-view + text heatmap	40	0.6818	0.4563

Teatime also benefits from the relation/region variant, reaching 0.5592 mIoU at unchanged localization accuracy. For Figurines, DINO cross-view with spatial text-heatmap distillation recovers the baseline localization score while improving mIoU from 0.4308 to 0.4343. Waldo Kitchen remains on the baseline checkpoint. Using the promoted Figurines, Ramen, and Teatime adaptor/cross-view checkpoints while keeping baseline Waldo Kitchen gives a candidate adaptor-enhanced macro score of 0.8712 LocAcc and 0.4979 mIoU.

The main failure mode is a localization/region tradeoff. The DINO relation and SAM3 region losses often smooth or broaden object-consistent heatmaps, which can raise thresholded overlap while moving the single argmax outside a small polygon. Because LERF-OVS LocAcc is an argmax-inside-mask metric, one feature-cell shift can hurt LocAcc even when the region response is visually cleaner. For submission, we therefore only promote adaptor checkpoints that preserve LocAcc; future training should add an explicit peak-preservation term or mask-aware small-object weighting before replacing the conservative main table.

Table 12 answers the cross-view question directly. DINO cross-view affinity and text-heatmap preservation can increase thresholded overlap, especially on Waldo at

Table 13: Frozen adaptor-space downstream probes on LERF-OVS. “Teacher” uses original RADIO RGB features; “Rendered” uses CTF-GS rendered features. Prototype segmentation uses same-category support masks from other frames. Matching uses the first annotated frame as source.

Adaptor	Probe	Teacher	
		LocAcc	mIoU
DINOv3	Prototype seg.	0.6543	0.0945
DINOv3	Source-target match	0.5957	0.1032
SAM3	Prototype seg.	0.8404	0.0757
SAM3	Source-target match	0.7872	0.0953

high temperature. The spatial heatmap variant recovers the Figurines localization score and becomes the only promoted cross-view LERF branch, but Waldo still loses the single-peak localization score. This suggests that cross-view affinity encourages coherent object regions while peak placement remains scene dependent; a full ProFuse-style pre-registration/context proposal module remains a future method extension rather than a paper-main result.

4.8 DINO/SAM Adaptor Downstream Probes

Table 13 is diagnostic rather than a claimed improvement: rendered features are competitive in DINOv3 mIoU but do not yet beat original RADIO features on aggregate, and unconstrained SAM3 prototype scoring exposes a larger teacher-rendered gap. Per-scene positives still exist, especially on Waldo Kitchen source-target matching, where rendered DINOv3 improves from 0.2500 to 0.5000 LocAcc and from 0.2708 to 0.2948 mIoU, and rendered SAM3 improves mIoU from 0.2965 to 0.3147 at unchanged LocAcc. These probes therefore support the feature-compatibility story while identifying adaptor-space preservation as the next optimization target. We therefore also report prompt-constrained task probes that better match the intended SAM3-adaptor and DINOv3-adaptor downstream use cases.

Table 14 separates two claims. First, rendered features improve prompt-constrained SAM3-adaptor mask mIoU over the frame-wise teacher for point prompts, box prompts, and mask propagation, while preserving perfect point-prompt localization. Second, source-background contrast makes DINOv3 mask propagation substantially stronger and gives rendered features higher propagation LocAcc than the teacher, but the rendered mask mIoU remains slightly lower. The higher rendered dense-matching similarity score suggests that the feature field can become over-smooth and select visually similar nearby regions. This motivates the cross-view DINO and text-heatmap protected training variants evaluated next.

Table 14: Promptable SAM3-adaptor and dense DINOv3-adaptor tasks on LERF-OVS. These probes use the same annotations as the grounding benchmark but replace text queries with adaptor-space prompts or source-target correspondences. SAM3 point/box prompts are prompt-constrained, mask propagation uses source-view background contrast weight of 0.5, and no external SAM3 mask decoder is called.

Task	Teacher		Rendered	
	LocAcc/Hit	mIoU/Score	LocAcc/Hit	mIoU/Score
SAM3 point prompt	1.0000	0.3700	1.0000	0.4169
SAM3 box prompt	0.8702	0.6560	0.8221	0.6638
SAM3 mask propagation	0.7872	0.3583	0.6667	0.3756
DINOv3 dense matching	0.5895	0.8543	0.5536	0.9048
DINOv3 mask propagation	0.7163	0.3921	0.7376	0.3684

Table 15: ScanNet v67 teacher-balanced direct point-query results.

Split	mIoU	mAcc
19 classes	0.3538	0.6076
15 classes	0.3573	0.6203
10 classes	0.4293	0.7051

4.9 ScanNet Direct Point-Query Transfer

Table 15 evaluates cross-domain feature usability on 10 ScanNet scenes using `gaussian_index` queries, `label_point` positions, and `label_index` opacity filtering. This is not presented as a full ScanNet semantic segmentation leaderboard result; it is a fair direct probe of the learned feature field.

Table 16 upgrades the earlier two-scene diagnostic to a full 10-scene ablation. DINO cross-view affinity improves the macro 19-class and 15-class probes more clearly than the 10-class probe, with positive macro mIoU and mAcc on all splits. The effect remains scene-dependent: the DINO branch improves 6/10, 7/10, and 5/10 scenes on the 19-, 15-, and 10-class mIoU splits, respectively. We therefore report it as a compatibility/FAC ablation rather than replacing the conservative main ScanNet row in Table 15.

4.10 Efficiency and Cost

Table 17 separates profiled evaluation cost from storage footprint accounting. The four LERF overlay evaluations take 124.770 seconds in total and peak at 2076 MiB, while the full 10-scene ScanNet v67 direct-query evaluation takes 150.903 seconds and peaks at 1666 MiB. Training throughput is kept as supplementary log-derived evidence because training wall time and explicit GPU telemetry are different measurement types.

Table 16: Full 10-scene ScanNet DINOv3 cross-view ablation. The branch is warm-started from the v67 teacher-balanced checkpoint and uses conservative DINO cross-view affinity weight $\lambda = 0.001$ with batch size 2.

Split	Base mIoU	DINO mIoU	Δ
19 classes	0.3538	0.3640	+0.0102
15 classes	0.3573	0.3662	+0.0089
10 classes	0.4293	0.4308	+0.0014

Table 17: Submission efficiency and cost evidence. Runtime rows use explicit freeze-profile telemetry; storage rows compare direct per-Gaussian 1280-D fp16 teacher features against the stored compact feature-field footprint.

Evidence	Scope	Cost
LERF eval	4 scenes	124.770 s / 31.193 s-scene
ScanNet eval	10 scenes	150.903 s / 15.090 s-scene
Storage	4 scenes	1.74 $\times$ –4.04 $\times$ saving

4.11 Qualitative Results

The current qualitative shortlist is stored in `output/radio_gs/reports/submission_freeze_figure_shortlist.txt`. The recommended main figure uses one frozen rendered grid from each LERF scene: Figurines frame 00152, Ramen frame 00024, Teatime frame 00140, and Waldo Kitchen frame 00089.

4.12 Failure Analysis

Table 18 turns the qualitative small-object observation into a paper-facing diagnostic. The lowest-ranked categories are mostly tiny or visually ambiguous objects, such as Figurines bags, character toys, and isolated kitchen tools. The failure pattern is not only low region overlap: some categories show nonzero mIoU but zero or partial LocAcc, which means the rendered heatmap covers a plausible object region while the single maximum lands outside the annotated polygon. This is why the paper treats mIoU improvements from adaptor or cross-view losses conservatively unless localization accuracy is preserved.

5 Discussion

Why feature reconstruction matters. The LERF-OVS results show that rendered RADIO-like features retain enough semantic structure for open-vocabulary localization. The ScanNet results suggest that the same representation can be probed outside the LERF object-query setting. The LERF direct-selection experiment further clarifies the boundary of this claim: the compact field supports direct primitive querying once rendered features

Table 18: Representative fragile LERF-OVS categories from the frozen rendered-feature evaluation. Low LocAcc with nonzero mIoU highlights the peak-vs-region trade-off.

Scene	DINO mAcc	Category	Δ	LocAcc	mIoU
Figurines	0.6205	bag	0.0128	0.00	0.00
Figurines	0.6313	miffy	0.0110	0.00	0.01
Teatime	0.7071	dalle	0.0020	0.00	0.01
Figurines		jake		0.00	0.02
Waldo Kitchen		ketchup		0.00	0.03
Waldo Kitchen		spatula		0.00	0.03
Ramen		corn		0.00	0.10
Figurines		red toy chair		0.00	0.13

are registered back to Gaussians, but the current voxel context is still weaker than explicit instance- or super-Gaussian grouping on fragmented scenes.

External-baseline scope. The LERF, LangSplat, and LEGaussians comparison rows are now traced to official paper or supplementary tables. They still come from cross-paper protocols, so the paper treats them as source-anchored context rows. A strict SOTA claim would re-run those baselines under the same evaluator and annotations used by RADIO-GS.

6 Limitations

CTF-GS depends on a pretrained Gaussian geometry backbone and does not claim to improve RGB reconstruction. Small objects remain challenging because the teacher feature resolution and rendered feature alignment limit heatmap sharpness. The current ScanNet protocol is a direct point-query feature probe, not a replacement for a full standardized segmentation benchmark. The registered LERF direct 3D object-selection readout substantially improves primitive querying and slightly exceeds the OpenGaussian official macro reference under the reported OpenGaussian-style context, but it remains weaker on Waldo Kitchen and is not a locally rerun same-evaluator comparison. The paper should therefore avoid claiming universal primitive-level SOTA. Finally, the external comparison is official-source but not reproduced under one evaluator, so strong SOTA-style claims require additional baseline reruns. The storage table reports checkpoint footprint rather than a deployed runtime memory allocator, so future packaging should separate feature-field parameters, rendering buffers, and optional downstream heads.

7 Conclusion

CTF-GS turns a 3D Gaussian scene into a reusable foundation-feature memory. By reconstructing frozen

Qualitative comparison on LERF-OVS

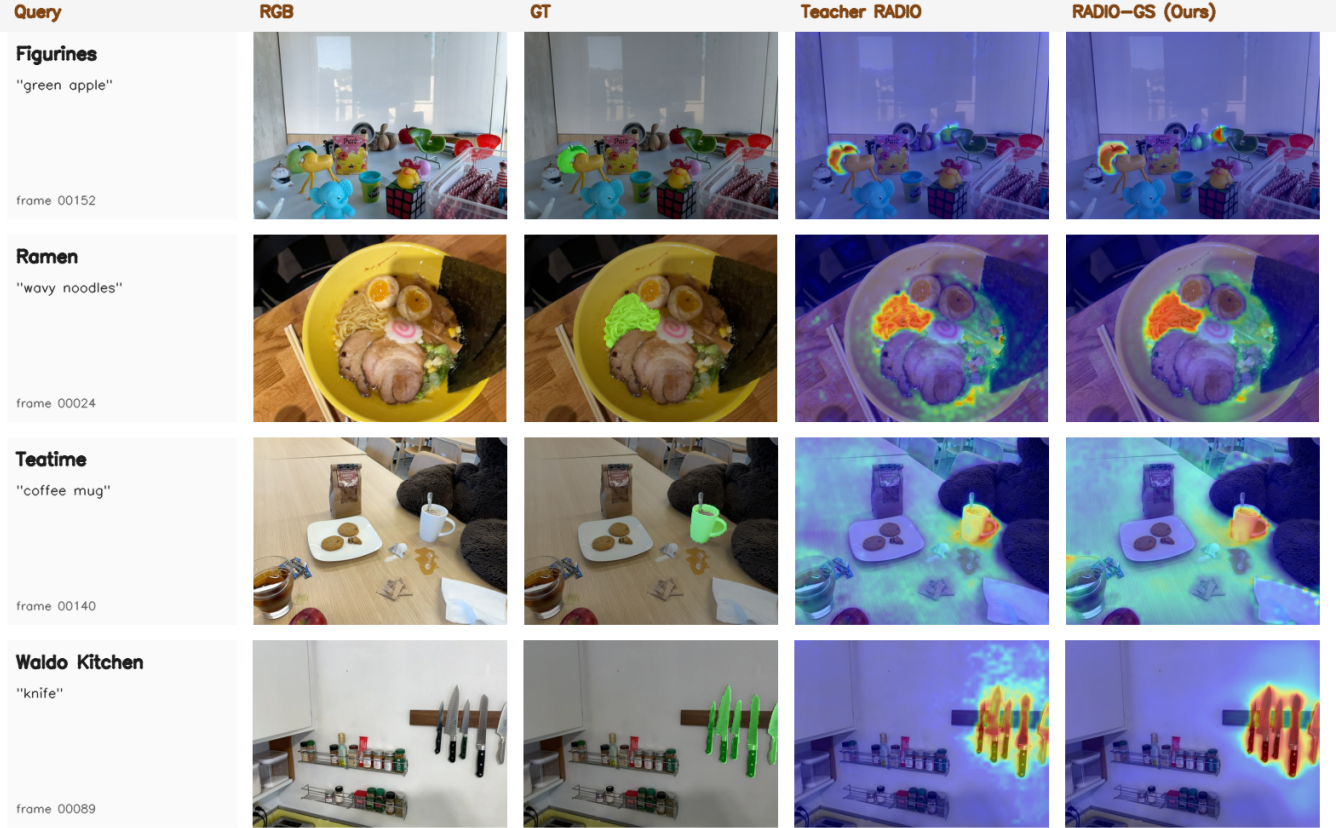


Figure 1: Qualitative open-vocabulary grounding examples from the frozen LERF overlay runs. Each row shows the RGB image, ground-truth mask overlay, frozen C-RADIOv4-H teacher heatmap, and rendered CTF-GS heatmap for a representative query.

RADIO features rather than training a scene-specific classifier, it supports open-vocabulary grounding and cross-domain feature queries from novel views. The current freeze package provides a coherent submission path: strong LERF-OVS internal results, fair ScanNet direct-query evidence, explicit storage and runtime profiles, failure analysis, and source-anchored external comparison caveats.

References

- [1] Jiazhong Cen, Jiemin Fang, Chen Yang, Lingxi Xie, Xiaopeng Zhang, Wei Shen, and Qi Tian. Segment any 3d gaussians. *arXiv preprint arXiv:2312.00860*, 2023.
- [2] Yen-Jen Chiou, Wei-Tse Cheng, and Yuan-Fu Yang. Profuse: Efficient cross-view context fusion for open-vocabulary 3d gaussian splatting. *arXiv preprint arXiv:2601.04754*, 2026.
- [3] Tianyu Huang, Runnan Chen, Dongting Hu, Fengming Huang, Mingming Gong, and Tongliang Liu. Openingsgaussian: Open-vocabulary instance gaussian segmentation with context-aware cross-view fusion. *arXiv preprint arXiv:2510.18253*, 2025.
- [4] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkuehler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 2023.
- [5] Justin Kerr, Chung Min Kim, Ken Goldberg, Angjoo Kanazawa, and Matthew Tancik. Lerf: Language embedded radiance fields. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.
- [6] Jun-Seong Kim, GeonU Kim, Yu-Ji Kim, Yu-Chiang Frank Wang, Jaesung Choe, and Tae-Hyun Oh. Dr. splat: Directly referring 3d gaussian splatting via direct language embedding registration. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.
- [7] Haijie Li, Yanmin Wu, Jiarui Meng, Qiankun Gao, Zhiyao Zhang, Ronggang Wang, and Jian Zhang. Instancegaussian: Appearance-semantic joint gaussian

- representation for 3d instance-level perception. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.
- [8] Kunyi Li, Michael Niemeyer, Sen Wang, Stefano Gasperini, Nassir Navab, and Federico Tombari. Opengaff: Open-vocabulary gaussian feature field with codebook attention. *arXiv preprint arXiv:2605.06088*, 2026.
- [9] Siyun Liang, Sen Wang, Kunyi Li, Michael Niemeyer, Stefano Gasperini, Hendrik P. A. Lensch, Nassir Navab, and Federico Tombari. Supergseg: Open-vocabulary 3d segmentation with structured super-gaussians. *arXiv preprint arXiv:2412.10231*, 2026.
- [10] Binh Long Nguyen, Kien Nguyen, Sridha Sridharan, Clinton Fookes, and Peyman Moghadam. Ilov3splat: Instance-level open-vocabulary 3d scene understanding in gaussian splatting. *arXiv preprint arXiv:2605.04506*, 2026.
- [11] NVIDIA. C-radiov4: Tech report. *arXiv preprint arXiv:2601.17237*, 2026.
- [12] Jens Piekenbrinck, Christian Schmidt, Alexander Hermans, Narunas Vaskevicius, Timm Linder, and Bastian Leibe. Opensplat3d: Open-vocabulary 3d instance segmentation using gaussian splatting. *arXiv preprint arXiv:2506.07697*, 2025.
- [13] Minghan Qin, Wanhua Li, Jiawei Zhou, Haoqian Wang, and Hanspeter Pfister. Langsplat: 3d language gaussian splatting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
- [14] Mike Ranzinger, Greg Heinrich, Jan Kautz, and Pavlo Molchanov. Am-radio: Agglomerative vision foundation model – reduce all domains into one. *arXiv preprint arXiv:2312.06709*, 2024.
- [15] Jin-Chuan Shi, Miao Wang, Hao-Bin Duan, and Shao-Hua Guan. Language embedded 3d gaussians for open-vocabulary scene understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
- [16] Wei Sun, Yanzhao Zhou, Jianbin Jiao, and Yuan Li. Cags: Open-vocabulary 3d scene understanding with context-aware gaussian splatting. *arXiv preprint arXiv:2504.11893*, 2025.
- [17] Yanmin Wu, Jiarui Meng, Haijie Li, Chenming Wu, Yahao Shi, Xinhua Cheng, Chen Zhao, Haocheng Feng, Errui Ding, Jingdong Wang, and Jian Zhang. Opengaussian: Towards point-level 3d gaussian-based open vocabulary understanding. *arXiv preprint arXiv:2406.02058*, 2024.
- [18] Xingxing Zuo, Pouya Samangouei, Yunwen Zhou, Yan Di, and Mingyang Li. Fmgs: Foundation model embedded 3d gaussian splatting for holistic 3d scene understanding. *arXiv preprint arXiv:2401.01970*, 2024.

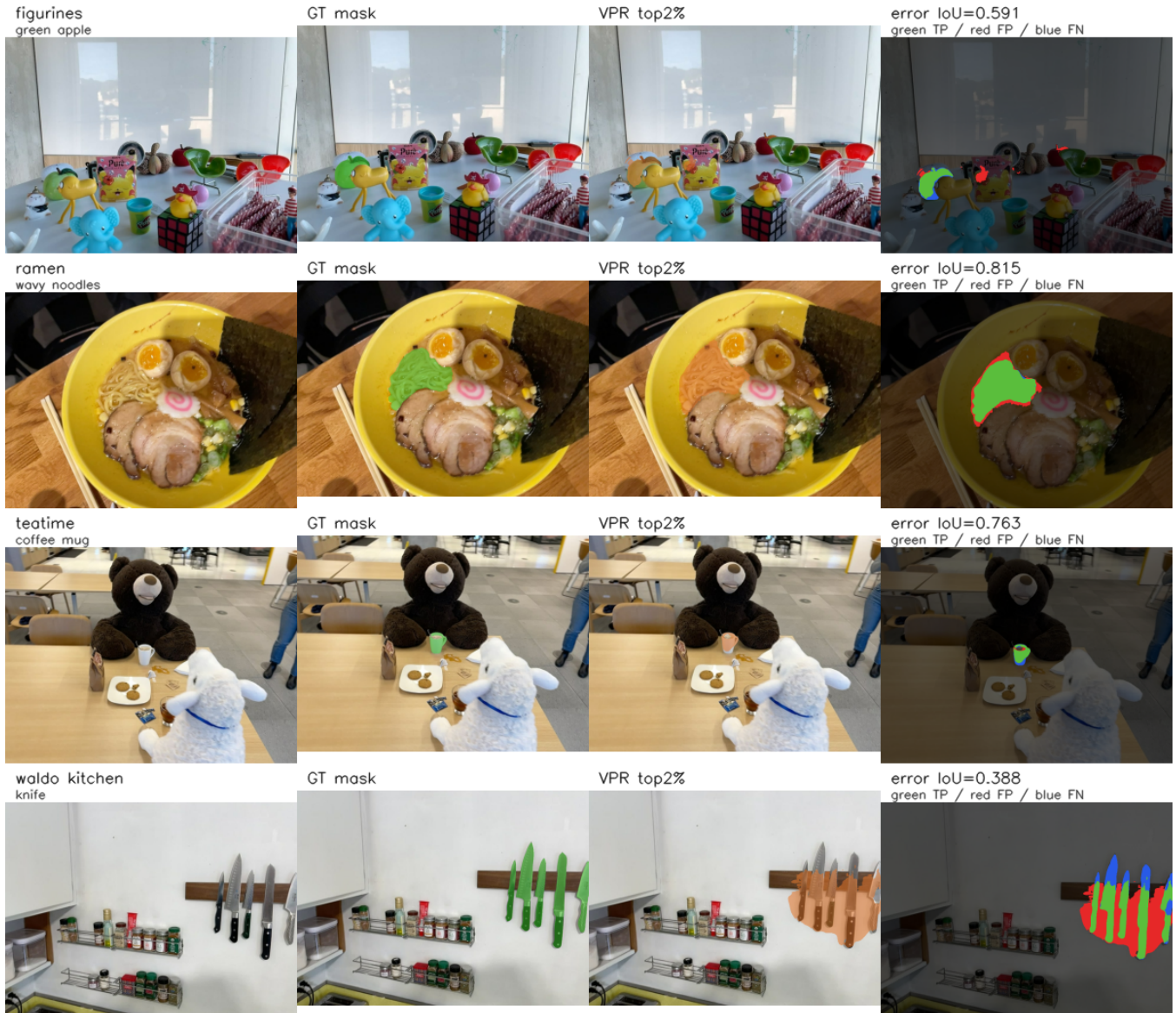


Figure 2: Qualitative LERF direct 3D object selection with VPR. Each row fixes a scene, annotated frame, and query, then shows RGB, ground-truth mask, the rendered mask from top-2% VPR-selected Gaussians, and an error map. Green is true positive, red is false positive, and blue is false negative. The Waldo Kitchen row illustrates the remaining fragmentation and clutter failure mode.

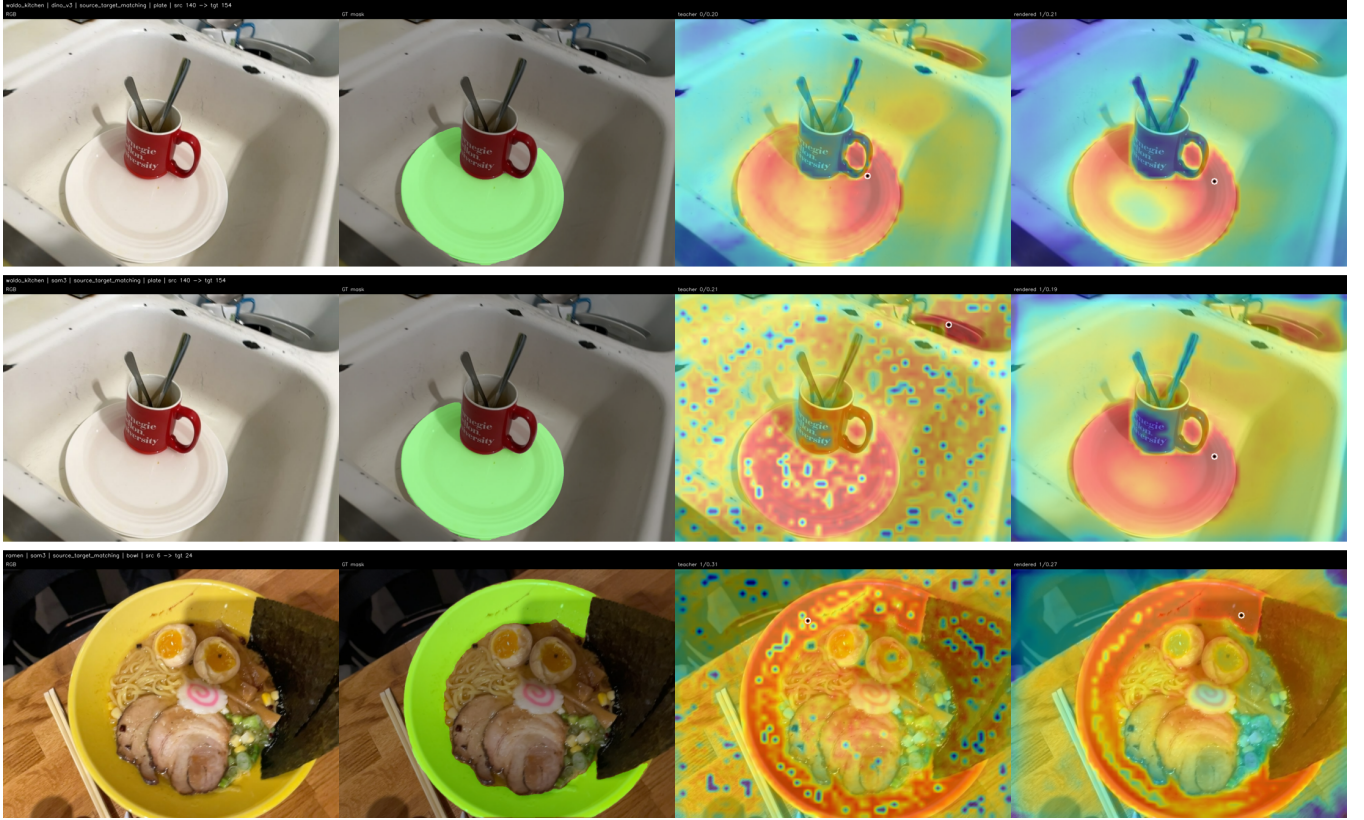


Figure 3: Qualitative DINOv3/SAM3 adaptor-space source-target matching. Each row shows RGB, GT mask, original RADIO teacher heatmap, and CTF-GS rendered heatmap. The Waldo Kitchen rows show cases where rendered features move the peak onto the target object; the Ramen row illustrates that the aggregate SAM3 gap remains visible even when localization is correct.

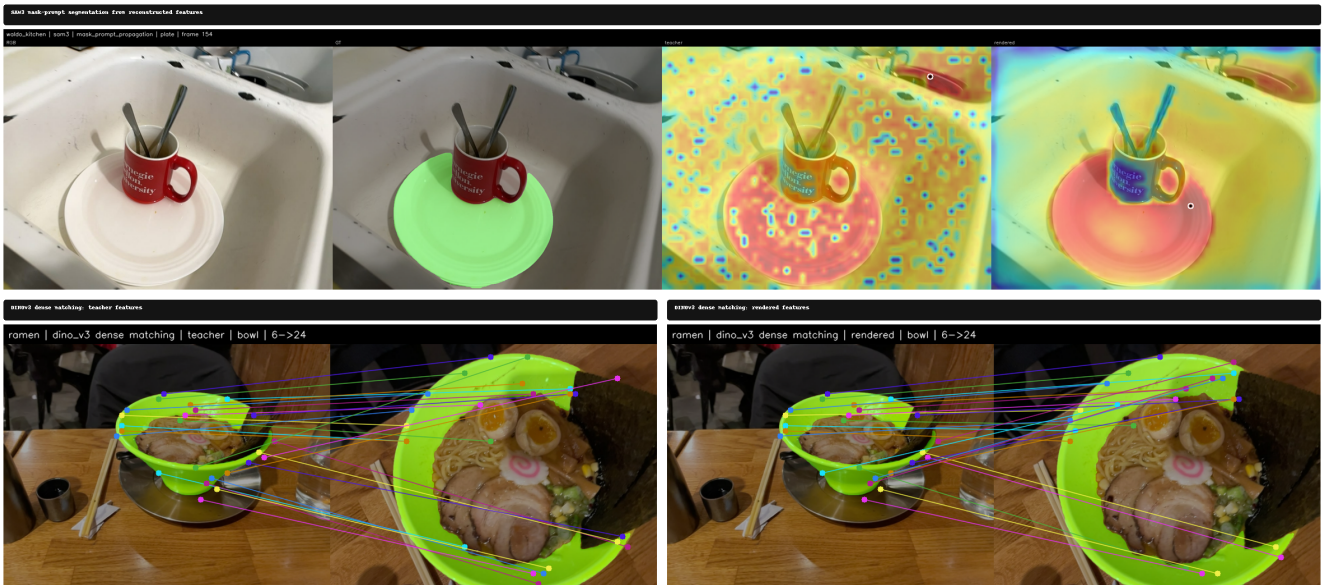


Figure 4: Qualitative promptable downstream probes from reconstructed features. Top: SAM3 mask-prompt propagation from a source annotation to a target view. Bottom: DINOv3 dense matching visualized with teacher and rendered correspondences.